

Energy Forecast System API Specification

1. Introduction

The Energy Forecast System API provides RESTful endpoints for authentication, dataset management, forecasting operations, accuracy evaluation, dashboard visualization, report generation, user access management, and forecasting parameter configuration.

The API architecture is derived from the system use case diagram, class diagram, and sequence diagrams. The API follows a layered architecture where endpoints interact with control classes responsible for business logic and entity classes responsible for data representation.

2. Base URL

```
/api
```

3. Authentication

Authentication is token-based.

Protected endpoints require:

```
Authorization: Bearer <token>
```

4. Authentication APIs

4.1 Sign In

Endpoint

```
POST /api/auth/login
```

Description

Authenticates a Facility Manager or System Administrator and creates an active session.

Request Body

```
{
  "username": "manager1",
  "password": "password123"
}
```

Success Response

```
{
  "message": "Login successful",
  "token": "jwt-token",
  "user": {
    "id": "USR001",
    "role": "FACILITY_MANAGER"
  }
}
```

Error Response

```
{
  "message": "Invalid credentials"
}
```

4.2 Sign Out

Endpoint

```
POST /api/auth/logout
```

Description

Terminates the active session of the authenticated user.

Headers

```
Authorization: Bearer <token>
```

Success Response

```
{
  "message": "Logout successful"
}
```

5. Dataset APIs

5.1 Upload Dataset

Endpoint

```
POST /api/datasets/upload
```

Description

Uploads a raw energy dataset into the system.

Headers

```
Authorization: Bearer <token>
Content-Type: multipart/form-data
```

Request Body

```
{
  "datasetName": "Building_A_Data",
  "file": "energy-data.csv"
}
```

Success Response

```
{
  "datasetId": "DS001",
  "status": "uploaded"
}
```

Error Response

```
{
  "message": "Dataset validation failed"
}
```

5.2 Run Preprocessing

Endpoint

```
POST /api/datasets/{datasetId}/preprocess
```

Description

Runs preprocessing on a previously uploaded dataset.

Success Response

```
{
  "datasetId": "DS001",
  "status": "processed"
}
```

5.3 Retrieve Dataset

Endpoint

```
GET /api/datasets/{datasetId}
```

Description

Retrieves dataset information and processing status.

Success Response

```
{
  "datasetId": "DS001",
  "status": "processed",
  "rawData": "...",
  "processedData": "..."
}
```

6. Forecast APIs

6.1 Run Forecast

Endpoint

```
POST /api/forecasts/run
```

Description

Executes the forecasting engine using a processed dataset and active configuration.

Request Body

```
{
  "datasetId": "DS001",
  "configurationId": "CFG001"
}
```

Success Response

```
{
  "forecastId": "FR001",
  "status": "generated"
}
```

6.2 Retrieve Forecast Result

Endpoint

```
GET /api/forecasts/{forecastId}
```

Description

Retrieves generated forecast values and metrics.

Success Response

```
{
  "forecastId": "FR001",
  "values": [],
}
```

```
"metrics": {}  
}
```

7. Accuracy Evaluation APIs

7.1 Evaluate Forecast Accuracy

Endpoint

```
POST /api/forecasts/{forecastId}/evaluate
```

Description

Computes forecast accuracy metrics against actual reference data.

Success Response

```
{  
  "forecastId": "FR001",  
  "metrics": {  
    "mae": 0.12,  
    "rmse": 0.18  
  }  
}
```

8. Dashboard APIs

8.1 View Dashboard

Endpoint

```
GET /api/dashboard
```

Description

Retrieves forecast summaries, metrics, and dashboard visualization data.

Success Response

```
{  
  "forecasts": [],
```

```
"metrics": [],  
"charts": []  
}
```

9. Report APIs

9.1 Export Report

Endpoint

```
GET /api/reports/export/{forecastId}
```

Description

Generates and exports a report for a selected forecast.

Success Response

```
{  
  "reportUrl": "/reports/FR001.pdf"  
}
```

10. User Management APIs

10.1 Create User

Endpoint

```
POST /api/users
```

Description

Creates a new user account.

Request Body

```
{  
  "username": "newmanager",  
  "password": "securepass",  
}
```

```
{
  "role": "FACILITY_MANAGER"
}
```

Success Response

```
{
  "message": "User created successfully"
}
```

10.2 Update User Permissions

Endpoint

```
PUT /api/users/{userId}/permissions
```

Description

Updates access permissions for a system user.

Request Body

```
{
  "role": "ADMINISTRATOR"
}
```

Success Response

```
{
  "message": "Permissions updated successfully"
}
```

10.3 Revoke User Access

Endpoint

```
DELETE /api/users/{userId}
```

Description

Revokes a user's access to the system.

Success Response

```
{
  "message": "User access revoked"
}
```

11. Forecast Configuration APIs

11.1 Update Forecast Parameters

Endpoint

```
PUT /api/configurations/{configId}
```

Description

Updates forecasting configuration parameters.

Request Body

```
{
  "seasonality": 12,
  "forecastHorizon": 24,
  "algorithm": "SARIMA"
}
```

Success Response

```
{
  "message": "Configuration updated"
}
```

12. HTTP Status Codes

Status Code	Meaning
200 OK	Request successful
201 Created	Resource created successfully
400 Bad Request	Invalid request data

Status Code	Meaning
401 Unauthorized	Authentication required
403 Forbidden	Access denied
404 Not Found	Resource not found
500 Internal Server Error	Server-side failure

13. Architectural Mapping

API Module	Main Control Class
Authentication	AuthenticationService
Dataset Management	DataIntakeService
Forecasting	ForecastEngine
Accuracy Evaluation	AccuracyEvaluator
Dashboard	DashboardView
Reporting	ReportGenerator
User Management	AccessController
Forecast Configuration	ForecastConfiguration

14. System Architecture Flow

```

Frontend Client
    ↓
REST API Controllers
    ↓
Control Classes / Services
    ↓
Entity Classes
    ↓
DatabaseConnector
    ↓
Database

```

The API layer serves as the communication interface between frontend clients and the system business logic. Control classes coordinate workflow execution, while entity classes represent persistent business data.